Lab 10
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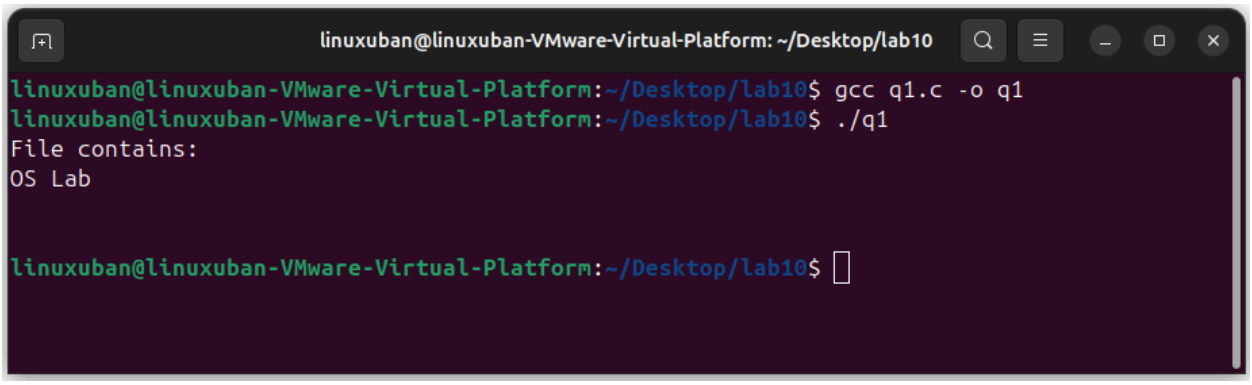
Read

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/mman.h>
#include <errno.h>

int main(int argc, char *argv[])
{
   int fd;
   char *data;

   if ((fd = open("m.c", O_RDONLY)) == -1)
   {
      perror("open");
      exit(1);
   }

   data = mmap(NULL, getpagesize(), PROT_READ, MAP_SHARED, fd, 0);

   printf("File contains: %s\n", data);

   return 0;
}
```

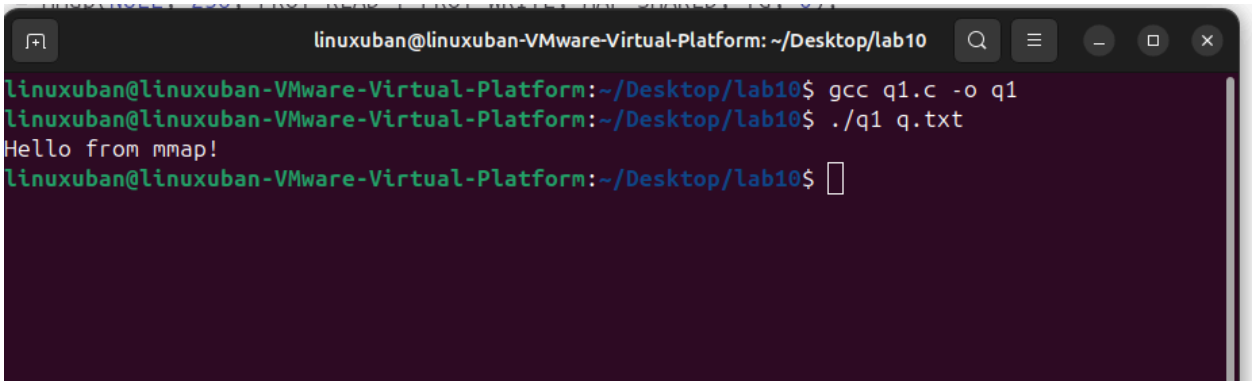
M.c
OS lab



Write

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/stat.h>
#include <sys/mman.h>

int main(int argc, char *argv[])
{
    int fd;
    void *file_memory;

    if (argc < 2) {
        printf("Usage: %s filename\n", argv[0]);
        return 1;
    }

    fd = open(argv[1], O_RDWR | O_CREAT, S_IRUSR | S_IWUSR);

    // Set file size to 256 bytes
    ftruncate(fd, 256);

    // Memory map
    file_memory = mmap(NULL, 256, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);

    if (file_memory == MAP_FAILED) {
        perror("mmap");
        return 1;
    }

    close(fd);

    // Write something into mapped memory
    sprintf((char *)file_memory, "Hello from mmap!\n");

    // Read it back
    printf("%s", (char *)file_memory);
```

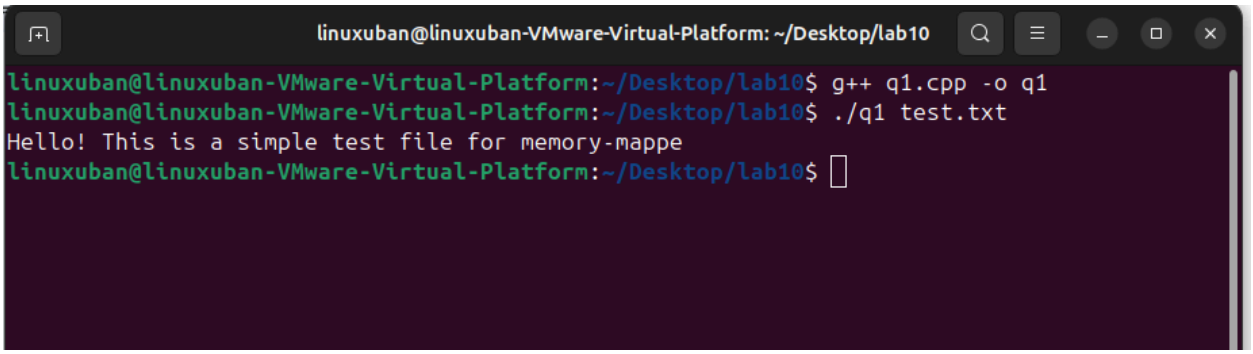
```
    // Unmap
    munmap(file_memory, 256);

    return 0;
}
```



Q1

```
#include <iostream>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>

using namespace std;

int main(int argc, char *argv[])
{
    int fd;
    struct stat sb;
    char *data;

    // Check command line argument
    if (argc < 2)
    {
        cout << "Usage: " << argv[0] << " <filename>" << endl;
        return 1;
    }
```

```
    // Open file
    fd = open(argv[1], O_RDONLY);
    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    // Get file size
    if (fstat(fd, &sb) == -1)
    {
        perror("fstat");
        close(fd);
        return 1;
    }

    // Memory map
    data = (char *)mmap(NULL, sb.st_size, PROT_READ, MAP_PRIVATE, fd, 0);
    if (data == MAP_FAILED)
    {
        perror("mmap");
        close(fd);
        return 1;
    }

    // Print first 50 characters (or less if file is smaller)
    int limit = (sb.st_size < 50) ? sb.st_size : 50;

    for (int i = 0; i < limit; i++)
    {
        cout << data[i];
    }
    cout << endl;

    // Unmap and close
    munmap(data, sb.st_size);
    close(fd);

    return 0;
}
```



Q2

```
#include <iostream>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <cstring>

using namespace std;

int main(int argc, char *argv[])
{
    if (argc < 2)
    {
        cout << "Usage: " << argv[0] << " <filename>" << endl;
        return 1;
    }

    // Open file with read + write permission
    int fd = open(argv[1], O_RDWR);
    if (fd == -1)
    {
        perror("open");
        return 1;
    }
```

```
    // Get file size
    struct stat sb;
    if (fstat(fd, &sb) == -1)
    {
        perror("fstat");
        close(fd);
        return 1;
    }

    if (sb.st_size < 10)
    {
        cout << "File too small to modify 10 characters" << endl;
        close(fd);
        return 1;
    }

    // Memory map with write access
    char *data = (char *)mmap(NULL, sb.st_size, PROT_READ | PROT_WRITE, MAP_SHARED,
fd, 0);
    if (data == MAP_FAILED)
    {
        perror("mmap");
        close(fd);
        return 1;
    }

    // Text to write (10 characters exactly)
    const char *text = "Hi There!!";

    // Copy into mapped memory
    memcpy(data, text, 10);

    // Unmap and close
    munmap(data, sb.st_size);
    close(fd);

    cout << "File modified successfully!" << endl; //First 10 characters

    return 0;
}
```


Open
test.txt
~/Desktop/lab10
Hello World! This is a test file.

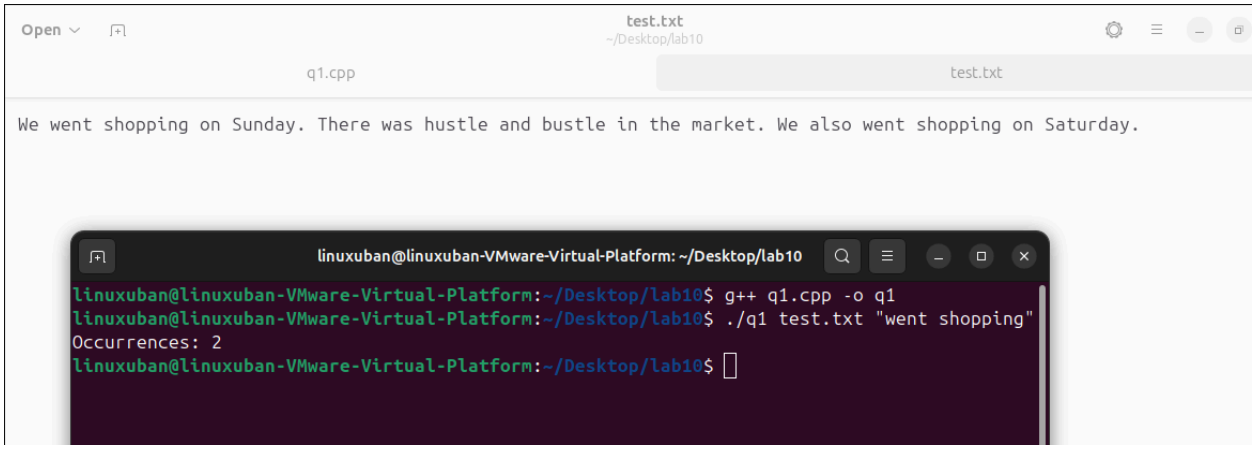

linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab10
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab10$ g++ q1.cpp -o q1
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab10$ ./q1 test.txt
File modified successfully!
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab10$


Open
test.txt
~/Desktop/lab10
Hi There!!d! This is a test file.

Q3

```
#include <iostream>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <cstring>

using namespace std;

int main(int argc, char *argv[])
{
    if (argc < 3)
    {
        cout << "Usage: " << argv[0] << " <filename> <substring>" << endl;
        return 1;
    }

    const char *filename = argv[1];
    const char *substr = argv[2];

    // Open file
    int fd = open(filename, O_RDONLY);
    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    // Get file size
    struct stat sb;
    if (fstat(fd, &sb) == -1)
    {
        perror("fstat");
        close(fd);
        return 1;
    }

    if (sb.st_size == 0)
    {
        cout << "File is empty" << endl;
        close(fd);
```

```
        return 1;
    }

    // Memory map
    char *data = (char *)mmap(NULL, sb.st_size, PROT_READ, MAP_PRIVATE, fd, 0);
    if (data == MAP_FAILED)
    {
        perror("mmap");
        close(fd);
        return 1;
    }

    int count = 0;
    int sub_len = strlen(substr);

    // Search substring
    for (int i = 0; i <= sb.st_size - sub_len; i++)
    {
        if (strncmp(&data[i], substr, sub_len) == 0)
        {
            count++;
        }
    }

    // Output result
    cout << "Occurrences: " << count << endl;

    // Cleanup
    munmap(data, sb.st_size);
    close(fd);

    return 0;
}
```


Open
test.txt
~/Desktop/lab10
q1.cpp
test.txt
We went shopping on Sunday. There was hustle and bustle in the market. We also went shopping on Saturday.
linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab10
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab10$ g++ q1.cpp -o q1
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab10$ ./q1 test.txt "went shopping"
Occurrences: 2
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab10$